

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2 *****
				3 *
				4 *
				5 * Miscellaneous-instruction-extensions facility 4 instruction tests
				6 * for RRE encoded:
				7 *
				8 * B968 CLZG - COUNT LEADING ZEROS
				9 * B969 CTZG - COUNT TRAILING ZEROS
				10 *
				11 * James Wekel March 2026
				12 *****
				14 *****
				15 *
				16 * basic instruction tests
				17 *
				18 *****
				19 * This program tests proper functioning of the Miscellaneous
				20 * Instruction Extensions facility 4 RRE Count Leading Zeros and
				21 * Count Trailing Zeros instructions.
				22 * Exceptions are not tested.
				23 *
				24 * PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
				25 * obvious coding errors. None of the tests are thorough. They are
				26 * NOT designed to test all aspects of any of the instructions.
				27 *
				28 *****
				29 *
				30 * *Testcase mie4-CLZG-CTZG
				31 * *
				32 * * Miscellaneous-instruction-extensions facility 4 instruction
				33 * * tests for RRE encoded:
				34 * *
				35 * * B968 CLZG - COUNT LEADING ZEROS
				36 * * B969 CTZG - COUNT TRAILING ZEROS
				37 * *
				38 * * # -----
				39 * * # This tests only the basic function of the instruction.
				40 * * # Exceptions are NOT tested.
				41 * * # -----
				42 * *
				43 * mainsize 2
				44 * numcpu 1
				45 * sysclear
				46 * archlvl z/Arch
				47 *
				48 * loadcore "\$(testpath)/mie4-CLZG-CTZG.core" 0x0
				49 *
				50 * diag8cmd enable # (needed for messages to Hercules console)
				51 * runtest 5 #
				52 * diag8cmd disable # (reset back to default)
				53 *
				54 * *Done
				55 *
				56 *****

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				58 *****
				59 * FCHECK Macro - Is a Facility Bit set?
				60 *
				61 * If the facility bit is NOT set, an message is issued and
				62 * the test is skipped.
				63 *
				64 * Fcheck uses R0, R1 and R2
				65 *
				66 * eg. FCHECK 134,'vector-packed-decimal'
				67 *****
				68 MACRO
				69 FCHECK &BITNO,&NOTSETMSG
				70 .* &BITNO : facility bit number to check
				71 .* &NOTSETMSG : 'facility name'
				72 LCLA &FBBYTE Facility bit in Byte
				73 LCLA &FBBIT Facility bit within Byte
				74
				75 LCLA &L(8)
				76 &L(1) SetA 128,64,32,16,8,4,2,1 bit positions within byte
				77
				78 &FBBYTE SETA &BITNO/8
				79 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)
				80 .* MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
				81
				82 B X&SYSNDX
				83 * Fcheck data area
				84 * skip messgae
				85 SKT&SYSNDX DC C' Skipping tests: '
				86 DC C&NOTSETMSG
				87 DC C' (bit &BITNO) is not installed.'
				88 SKL&SYSNDX EQU *-SKT&SYSNDX
				89 * facility bits
				90 DS FD gap
				91 FB&SYSNDX DS 4FD
				92 DS FD gap
				93 *
				94 X&SYSNDX EQU *
				95 LA R0,((X&SYSNDX-FB&SYSNDX)/8)-1
				96 STFLE FB&SYSNDX get facility bits
				97
				98 XGR R0,R0
				99 IC R0,FB&SYSNDX+&FBBYTE get fbit byte
				100 N R0,=F'&FBBIT' is bit set?
				101 BNZ XC&SYSNDX
				102 *
				103 * facility bit not set, issue message and exit
				104 *
				105 LA R0,SKL&SYSNDX message length
				106 LA R1,SKT&SYSNDX message address
				107 BAL R2,MSG
				108
				109 B EOJ
				110 XC&SYSNDX EQU *
				111 MEND

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				149	*****
				150	* Low core PSWs
				151	*****
00000000		00000000	000016AF	152	MIE4TST START 0
		00000000		153	USING MIE4TST,R0 Low core addressability
		00000140	00000000	154	
				155	SVOLDPSW EQU MIE4TST+X'140' z/Arch Supervisor call old PSW
00000000		00000000	000001A0	157	ORG MIE4TST+X'1A0' z/Architecure RESTART PSW
000001A0	00000001 80000000			158	DC X'0000000180000000'
000001A8	00000000 00000200			159	DC AD(BEGIN)
000001B0		000001B0	000001D0	161	ORG MIE4TST+X'1D0' z/Architecure PROGRAM CHECK PSW
000001D0	00020001 80000000			162	DC X'0002000180000000'
000001D8	00000000 0000DEAD			163	DC AD(X'DEAD')
000001E0		000001E0	00000200	165	ORG MIE4TST+X'200' Start of actual test program...
				167	*****
				168	* The actual "MIE4TST" program itself...
				169	*****
				170	* Architecture Mode: z/Arch
				171	* Register Usage:
				172	
				173	* R0 (work)
				174	* R1-4 (work)
				175	* R5 Testing control table - current test base
				176	* R6-R7 (work)
				177	* R8 First base register
				178	* R9 Second base register
				179	* R10 Third base register
				180	* R11 M4TEST call return
				181	* R12 M4TESTS register
				182	* R13 (work)
				183	* R14 Subroutine call
				184	* R15 Secondary Subroutine call or work
				185	* R15
				186	*
				187	*****
00000200		00000200		189	USING BEGIN,R8 FIRST Base Register
00000200		00001200		190	USING BEGIN+4096,R9 SECOND Base Register
00000200		00002200		191	USING BEGIN+8192,R10 THIRD Base Register
00000200	0580			193	BEGIN BALR R8,0 Initalize FIRST base register
00000202	0680			194	BCTR R8,0 Initalize FIRST base register
00000204	0680			195	BCTR R8,0 Initalize FIRST base register
00000206	4190 8800		00000800	197	LA R9,2048(,R8) Initalize SECOND base register
0000020A	4190 9800		00000800	198	LA R9,2048(,R9) Initalize SECOND base register
				199	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				367 *****
				368 * Normal completion or Abnormal termination PSWs
				369 *****
00000458	00020001 80000000			371 EOJPSW DC 0D'0',X'0002000180000000',AD(0)
00000468	B2B2 8258		00000458	373 EOJ LPSWE EOJPSW Normal completion
00000470	00020001 80000000			375 FAILPSW DC 0D'0',X'0002000180000000',AD(X'BAD')
00000480	B2B2 8270		00000470	377 FAILTEST LPSWE FAILPSW Abnormal termination
				379 *****
				380 * Working Storage
				381 *****
00000484	00000000			383 CTLR0 DS F CR0
00000488	00000000			384 DS F
0000048C				386 LTORG , Literals pool
0000048C	00000008			387 =F'8'
00000490	00001668			388 =A(M4TESTS)
00000494	00000001			389 =F'1'
00000498	0000			390 =H'0'
0000049A	005F			391 =AL2(L'MSGMSG)
				392
				393 * some constants
				394
	00000400	00000001		395 K EQU 1024 One KB
	00001000	00000001		396 PAGE EQU (4*K) Size of one page
	00010000	00000001		397 K64 EQU (64*K) 64 KB
	00100000	00000001		398 MB EQU (K*K) 1 MB
				399
	AABBCCDD	00000001		400 REG2PATT EQU X'AABBCCDD' Polluted Register pattern
	000000DD	00000001		401 REG2LOW EQU X'DD' (last byte above)

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT
					441 *****
					442 * M4TEST DSECT
					443 *****
					445 M4TEST DSECT ,
00000000	00000000				446 TSUB DC A(0) pointer to test
00000004	0000				447 TNUM DC H'00' Test Number
00000006	00				448 DC X'00'
					449
00000007	40404040	40404040			450 OPNAME DC CL8' ' E7 name
00000010	00000000				451 R2ADDR DC A(0) address of R2 source
00000014	00000000				452 R3ADDR DC A(0) address of R3 source
00000018	00000000				453 RELEN DC A(0) RESULT LENGTH
0000001C	00000000				454 READDR DC A(0) result (expected) address
00000020	00000000	00000000			455 DS FD gap
00000028	00000000	00000000			456 R1OUTPUT DS D r1 Output
00000030	00000000	00000000			457 DS FD gap
					458
					459 * test routine will be here (from RRE macro)
					460 *
					461 * followed by
					462 * EXPECTED RESULT
			00000000	000016AF	464 MIE4TST CSECT ,
000010A8					465 DS 0F
					467 *****
					468 * Macros to help build test tables
					469 *****
					471 *
					472 * macro to generate individual test
					473 *
					474 MACRO
					475 RRE &INST
					476 .* &INST - RRE instruction under test
					477
					478
					479 GBLA &TNUM
			&TNUM		480 SETA &TNUM+1
					481
					482 DS 0FD
					483 USING *,R5 base for test data and test routine
					484
			T&TNUM		485 DC A(X&TNUM) address of test routine
					486 DC H'&TNUM' test number
					487 DC X'00'
					488 DC CL8'&INST' instruction name
					489 DC A(RE&TNUM+8) address of R2 source
					490 DC A(RE&TNUM+16) address of R3 source
					491 DC A(8) result length

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT			
					492	REA&TNUM	DC	A(RE&TNUM)result address
					493		DS	FDgap
					494	R10&TNUM	DS	D
					495		DS	FD
					496	.*		
					497	*		
					498	X&TNUM	DS	0F
					499		LG	R1,R1FUDGE
					500			
					501		XGR	R2,R2
					502		L	R2,R2ADDRload r2 source address
					503		LG	R2,0(R2)load source
					504			
					505		&INST	test instruction
					506		STG	R1,R10&TNUMsave R1 output
					507			
					508		BR	R11return
					509			
					510	RE&TNUM	DC	0Fxl16 expected result
					511			
					512		DROP	R5
					513		MEND	
					515	*		
					516	* macro to generate table of pointers to individual tests		
					517	*		
					518		MACRO	
					519		PTTABLE	
					520		GBLA	&TNUM
					521		LCLA	&CUR
					522	&CUR	SETA	1
					523	.*		
					524	TTABLE	DS	0F
					525	. LOOP	ANOP	
					526	.*		
					527		DC	A(T&CUR)
					528	.*		
					529	&CUR	SETA	&CUR+1
					530		AIF	(&CUR LE &TNUM). LOOP
					531	*		
					532		DC	A(0)END OF TABLE
					533		DC	A(0)
					534	.*		
					535		MEND	
					536			

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				538 *****	
				539 * MIE4 RRE tests	
				540 *****	
000010A8	00000000 00000000			541 PRINT DATA	
				542 DS FD	
				543 *	
				544 * B968 CLZG - COUNT LEADING ZEROS	
				545 * B969 CTZG - COUNT TRAILING ZEROS	
				546 *	
				547 * RRE instruction	
				548 * followed by	
				549 * 8 byte expected result (R1)	
				550 * 8 byte R2 source	
				551 *-----	
				552 * CLZG - COUNT LEADING ZEROS	
				553 *-----	
				554 *	
000010B0				555 RRE CLZG	
000010B0		000010B0		556+ DS 0FD	
000010B0	000010E8			557+ USING *,R5	base for test data and test routine
000010B4	0001			558+T1 DC A(X1)	address of test routine
000010B6	00			559+ DC H'1'	test number
000010B7	C3D3E9C7 40404040			560+ DC X'00'	
000010C0	00001110			561+ DC CL8'CLZG'	instruction name
000010C4	00001118			562+ DC A(RE1+8)	address of R2 source
000010C8	00000008			563+ DC A(RE1+16)	address of R3 source
000010CC	00001108			564+ DC A(8)	result length
000010D0	00000000 00000000			565+REA1 DC A(RE1)	result address
000010D8	00000000 00000000			566+ DS FD	gap
000010E0	00000000 00000000			567+R101 DS D	R1 output
				568+ DS FD	gap
				569+*	
000010E8				570+X1 DS 0F	
000010E8	E310 8E88 0004	00001088		571+ LG R1,R1FUDGE	
000010EE	B982 0022			572+ XGR R2,R2	
000010F2	5820 5010	00000010		573+ L R2,R2ADDR	load r2 source address
000010F6	E322 0000 0004	00000000		574+ LG R2,0(R2)	load source
000010FC				577+ DS 0H	
000010FC	B9680012			578+ DC XL4'B9680012'	CLZQ r1,r2
00001100	E310 5028 0024	000010D8		580+ STG R1,R101	save R1 output
00001106	07FB			581+ BR R11	return
00001108				582+RE1 DC 0F	xl16 expected result
00001108				583+ DROP R5	
00001108	00000000 00000000			584 DC XL8'0000 0000 0000 0000'	result
00001110	AAAAAAAA AAAAAAAAAA			585 DC XL8'AAAA AAAA AAAA AAAA'	R2
				586	
00001118				587 RRE CLZG	
00001118		00001118		588+ DS 0FD	
00001118	00001150			589+ USING *,R5	base for test data and test routine
0000111C	0002			590+T2 DC A(X2)	address of test routine
0000111E	00			591+ DC H'2'	test number
0000111F	C3D3E9C7 40404040			592+ DC X'00'	
00001128	00001178			593+ DC CL8'CLZG'	instruction name
				594+ DC A(RE2+8)	address of R2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000112C	00001180			595+	DC	A(RE2+16)	address of R3 source
00001130	00000008			596+	DC	A(8)	result length
00001134	00001170			597+REA2	DC	A(RE2)	result address
00001138	00000000 00000000			598+	DS	FD	gap
00001140	00000000 00000000			599+R102	DS	D	R1 output
00001148	00000000 00000000			600+	DS	FD	gap
				601+*			
00001150				602+X2	DS	0F	
00001150	E310 8E88 0004		00001088	603+	LG	R1,R1FUDGE	
00001156	B982 0022			604+	XGR	R2,R2	
0000115A	5820 5010		00000010	605+	L	R2,R2ADDR	load r2 source address
0000115E	E322 0000 0004		00000000	606+	LG	R2,0(R2)	load source
00001164				609+	DS	0H	
00001164	B9680012			610+	DC	XL4'B9680012'	CLZQ r1,r2
00001168	E310 5028 0024		00001140	612+	STG	R1,R102	save R1 output
0000116E	07FB			613+	BR	R11	return
00001170				614+RE2	DC	0F	xl16 expected result
00001170				615+	DROP	R5	
00001170	00000000 00000004			616	DC	XL8'0000 0000 0000 0004'	result
00001178	0AAAAAAA AAAAAAAA			617	DC	XL8'0AAA AAAA AAAA AAAA'	R2
				618			
00001180				619	RRE	CLZG	
00001180		00001180		620+	DS	0FD	
00001180	000011B8			621+	USING	*,R5	base for test data and test routine
00001184	0003			622+T3	DC	A(X3)	address of test routine
00001186	00			623+	DC	H'3'	test number
00001186	00			624+	DC	X'00'	
00001187	C3D3E9C7 40404040			625+	DC	CL8'CLZG'	instruction name
00001190	000011E0			626+	DC	A(RE3+8)	address of R2 source
00001194	000011E8			627+	DC	A(RE3+16)	address of R3 source
00001198	00000008			628+	DC	A(8)	result length
0000119C	000011D8			629+REA3	DC	A(RE3)	result address
000011A0	00000000 00000000			630+	DS	FD	gap
000011A8	00000000 00000000			631+R103	DS	D	R1 output
000011B0	00000000 00000000			632+	DS	FD	gap
				633+*			
000011B8				634+X3	DS	0F	
000011B8	E310 8E88 0004		00001088	635+	LG	R1,R1FUDGE	
000011BE	B982 0022			636+	XGR	R2,R2	
000011C2	5820 5010		00000010	637+	L	R2,R2ADDR	load r2 source address
000011C6	E322 0000 0004		00000000	638+	LG	R2,0(R2)	load source
000011CC				641+	DS	0H	
000011CC	B9680012			642+	DC	XL4'B9680012'	CLZQ r1,r2
000011D0	E310 5028 0024		000011A8	644+	STG	R1,R103	save R1 output
000011D6	07FB			645+	BR	R11	return
000011D8				646+RE3	DC	0F	xl16 expected result
000011D8				647+	DROP	R5	
000011D8	00000000 00000006			648	DC	XL8'0000 0000 0000 0006'	result
000011E0	03AAAAAA AAAAAAAA			649	DC	XL8'03AA AAAA AAAA AAAA'	R2
				650			
000011E8				651	RRE	CLZG	
				652+	DS	0FD	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000011E8		000011E8		653+	USING *,R5	base for test data and test routine
000011E8	00001220			654+T4	DC A(X4)	address of test routine
000011EC	0004			655+	DC H'4'	test number
000011EE	00			656+	DC X'00'	
000011EF	C3D3E9C7 40404040			657+	DC CL8'CLZG'	instruction name
000011F8	00001248			658+	DC A(RE4+8)	address of R2 source
000011FC	00001250			659+	DC A(RE4+16)	address of R3 source
00001200	00000008			660+	DC A(8)	result length
00001204	00001240			661+REA4	DC A(RE4)	result address
00001208	00000000 00000000			662+	DS FD	gap
00001210	00000000 00000000			663+R104	DS D	R1 output
00001218	00000000 00000000			664+	DS FD	gap
				665+*		
00001220				666+X4	DS 0F	
00001220	E310 8E88 0004		00001088	667+	LG R1,R1FUDGE	
00001226	B982 0022			668+	XGR R2,R2	
0000122A	5820 5010		00000010	669+	L R2,R2ADDR	load r2 source address
0000122E	E322 0000 0004		00000000	670+	LG R2,0(R2)	load source
00001234				673+	DS 0H	
00001234	B9680012			674+	DC XL4'B9680012'	CLZQ r1,r2
00001238	E310 9010 0024		00001210	676+	STG R1,R104	save R1 output
0000123E	07FB			677+	BR R11	return
00001240				678+RE4	DC 0F	xl16 expected result
00001240				679+	DROP R5	
00001240	00000000 00000010			680	DC XL8'0000 0000 0000 0010'	result
00001248	0000AAAA AAAAAAAA			681	DC XL8'0000 AAAA AAAA AAAA'	R2
				682		
				683	RRE CLZG	
00001250		00001250		684+	DS 0FD	
00001250				685+	USING *,R5	base for test data and test routine
00001250	00001288			686+T5	DC A(X5)	address of test routine
00001254	0005			687+	DC H'5'	test number
00001256	00			688+	DC X'00'	
00001257	C3D3E9C7 40404040			689+	DC CL8'CLZG'	instruction name
00001260	000012B0			690+	DC A(RE5+8)	address of R2 source
00001264	000012B8			691+	DC A(RE5+16)	address of R3 source
00001268	00000008			692+	DC A(8)	result length
0000126C	000012A8			693+REA5	DC A(RE5)	result address
00001270	00000000 00000000			694+	DS FD	gap
00001278	00000000 00000000			695+R105	DS D	R1 output
00001280	00000000 00000000			696+	DS FD	gap
				697+*		
00001288				698+X5	DS 0F	
00001288	E310 8E88 0004		00001088	699+	LG R1,R1FUDGE	
0000128E	B982 0022			700+	XGR R2,R2	
00001292	5820 5010		00000010	701+	L R2,R2ADDR	load r2 source address
00001296	E322 0000 0004		00000000	702+	LG R2,0(R2)	load source
0000129C				705+	DS 0H	
0000129C	B9680012			706+	DC XL4'B9680012'	CLZQ r1,r2
000012A0	E310 5028 0024		00001278	708+	STG R1,R105	save R1 output
000012A6	07FB			709+	BR R11	return
000012A8				710+RE5	DC 0F	xl16 expected result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT						
000012A8				711+	DROP	R5				
000012A8	00000000 00000018			712	DC	XL8'0000 0000 0000 0018'	result			
000012B0	000000AA AAAAAAAA			713	DC	XL8'0000 00AA AAAA AAAA'	R2			
				714						
				715	RRE	CLZG				
000012B8				716+	DS	0FD				
000012B8		000012B8		717+	USING	*,R5	base for test data and test routine			
000012B8	000012F0			718+T6	DC	A(X6)	address of test routine			
000012BC	0006			719+	DC	H'6'	test number			
000012BE	00			720+	DC	X'00'				
000012BF	C3D3E9C7 40404040			721+	DC	CL8'CLZG'	instruction name			
000012C8	00001318			722+	DC	A(RE6+8)	address of R2 source			
000012CC	00001320			723+	DC	A(RE6+16)	address of R3 source			
000012D0	00000008			724+	DC	A(8)	result length			
000012D4	00001310			725+REA6	DC	A(RE6)	result address			
000012D8	00000000 00000000			726+	DS	FD	gap			
000012E0	00000000 00000000			727+R106	DS	D	R1 output			
000012E8	00000000 00000000			728+	DS	FD	gap			
				729+*						
000012F0				730+X6	DS	0F				
000012F0	E310 8E88 0004		00001088	731+	LG	R1,R1FUDGE				
000012F6	B982 0022			732+	XGR	R2,R2				
000012FA	5820 5010		00000010	733+	L	R2,R2ADDR	load r2 source address			
000012FE	E322 0000 0004		00000000	734+	LG	R2,0(R2)	load source			
00001304				737+	DS	0H				
00001304	B9680012			738+	DC	XL4'B9680012'	CLZQ r1,r2			
00001308	E310 5028 0024		000012E0	740+	STG	R1,R106	save R1 output			
0000130E	07FB			741+	BR	R11	return			
00001310				742+RE6	DC	0F	xl16 expected result			
00001310				743+	DROP	R5				
00001310	00000000 00000030			744	DC	XL8'0000 0000 0000 0030'	result			
00001318	00000000 0000AAAA			745	DC	XL8'0000 0000 0000 AAAA'	R2			
				746						
				747	RRE	CLZG				
00001320				748+	DS	0FD				
00001320		00001320		749+	USING	*,R5	base for test data and test routine			
00001320	00001358			750+T7	DC	A(X7)	address of test routine			
00001324	0007			751+	DC	H'7'	test number			
00001326	00			752+	DC	X'00'				
00001327	C3D3E9C7 40404040			753+	DC	CL8'CLZG'	instruction name			
00001330	00001380			754+	DC	A(RE7+8)	address of R2 source			
00001334	00001388			755+	DC	A(RE7+16)	address of R3 source			
00001338	00000008			756+	DC	A(8)	result length			
0000133C	00001378			757+REA7	DC	A(RE7)	result address			
00001340	00000000 00000000			758+	DS	FD	gap			
00001348	00000000 00000000			759+R107	DS	D	R1 output			
00001350	00000000 00000000			760+	DS	FD	gap			
				761+*						
00001358				762+X7	DS	0F				
00001358	E310 8E88 0004		00001088	763+	LG	R1,R1FUDGE				
0000135E	B982 0022			764+	XGR	R2,R2				
00001362	5820 5010		00000010	765+	L	R2,R2ADDR	load r2 source address			
00001366	E322 0000 0004		00000000	766+	LG	R2,0(R2)	load source			

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000136C				769+	DS	0H	
0000136C	B9680012			770+	DC	XL4'B9680012'	CLZQ r1,r2
00001370	E310 5028 0024		00001348	772+	STG	R1,R107	save R1 output
00001376	07FB			773+	BR	R11	return
00001378				774+RE7	DC	0F	xl16 expected result
00001378				775+	DROP	R5	
00001378	00000000 00000040			776	DC	XL8'0000 0000 0000 0040'	result
00001380	00000000 00000000			777	DC	XL8'0000 0000 0000 0000'	R2
				778			
				779 *			
				780 *	CTZG	- COUNT TRAILING ZEROS	
				781 *			
				782 *			
				783	RRE	CTZG	
00001388				784+	DS	0FD	
00001388		00001388		785+	USING	*,R5	base for test data and test routine
00001388	000013C0			786+T8	DC	A(X8)	address of test routine
0000138C	0008			787+	DC	H'8'	test number
0000138E	00			788+	DC	X'00'	
0000138F	C3E3E9C7 40404040			789+	DC	CL8'CTZG'	instruction name
00001398	000013E8			790+	DC	A(RE8+8)	address of R2 source
0000139C	000013F0			791+	DC	A(RE8+16)	address of R3 source
000013A0	00000008			792+	DC	A(8)	result length
000013A4	000013E0			793+REA8	DC	A(RE8)	result address
000013A8	00000000 00000000			794+	DS	FD	gap
000013B0	00000000 00000000			795+R108	DS	D	R1 output
000013B8	00000000 00000000			796+	DS	FD	gap
				797+*			
000013C0				798+X8	DS	0F	
000013C0	E310 8E88 0004		00001088	799+	LG	R1,R1FUDGE	
000013C6	B982 0022			800+	XGR	R2,R2	
000013CA	5820 5010		00000010	801+	L	R2,R2ADDR	load r2 source address
000013CE	E322 0000 0004		00000000	802+	LG	R2,0(R2)	load source
000013D4				805+	DS	0H	
000013D4	B9690012			806+	DC	XL4'B9690012'	CTZQ r1,r2
000013D8	E310 5028 0024		000013B0	808+	STG	R1,R108	save R1 output
000013DE	07FB			809+	BR	R11	return
000013E0				810+RE8	DC	0F	xl16 expected result
000013E0				811+	DROP	R5	
000013E0	00000000 00000000			812	DC	XL8'0000 0000 0000 0000'	result
000013E8	55555555 55555555			813	DC	XL8'5555 5555 5555 5555'	R2
				814			
				815	RRE	CTZG	
000013F0				816+	DS	0FD	
000013F0		000013F0		817+	USING	*,R5	base for test data and test routine
000013F0	00001428			818+T9	DC	A(X9)	address of test routine
000013F4	0009			819+	DC	H'9'	test number
000013F6	00			820+	DC	X'00'	
000013F7	C3E3E9C7 40404040			821+	DC	CL8'CTZG'	instruction name
00001400	00001450			822+	DC	A(RE9+8)	address of R2 source
00001404	00001458			823+	DC	A(RE9+16)	address of R3 source
00001408	00000008			824+	DC	A(8)	result length
0000140C	00001448			825+REA9	DC	A(RE9)	result address

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001410	00000000 00000000			826+	DS	FD	gap
00001418	00000000 00000000			827+R109	DS	D	R1 output
00001420	00000000 00000000			828+	DS	FD	gap
				829+*			
00001428				830+X9	DS	0F	
00001428	E310 8E88 0004		00001088	831+	LG	R1,R1FUDGE	
0000142E	B982 0022			832+	XGR	R2,R2	
00001432	5820 5010		00000010	833+	L	R2,R2ADDR	load r2 source address
00001436	E322 0000 0004		00000000	834+	LG	R2,0(R2)	load source
0000143C				837+	DS	0H	
0000143C	B9690012			838+	DC	XL4'B9690012'	CTZQ r1,r2
00001440	E310 5028 0024		00001418	840+	STG	R1,R109	save R1 output
00001446	07FB			841+	BR	R11	return
00001448				842+RE9	DC	0F	xl16 expected result
00001448				843+	DROP	R5	
00001448	00000000 00000004			844	DC	XL8'0000 0000 0000 0004'	result
00001450	0A555555 55555550			845	DC	XL8'0A55 5555 5555 5550'	R2
				846			
				847	RRE	CTZG	
00001458				848+	DS	0FD	
00001458		00001458		849+	USING	*,R5	base for test data and test routine
00001458	00001490			850+T10	DC	A(X10)	address of test routine
0000145C	000A			851+	DC	H'10'	test number
0000145E	00			852+	DC	X'00'	
0000145F	C3E3E9C7 40404040			853+	DC	CL8'CTZG'	instruction name
00001468	000014B8			854+	DC	A(RE10+8)	address of R2 source
0000146C	000014C0			855+	DC	A(RE10+16)	address of R3 source
00001470	00000008			856+	DC	A(8)	result length
00001474	000014B0			857+REA10	DC	A(RE10)	result address
00001478	00000000 00000000			858+	DS	FD	gap
00001480	00000000 00000000			859+R1010	DS	D	R1 output
00001488	00000000 00000000			860+	DS	FD	gap
				861+*			
00001490				862+X10	DS	0F	
00001490	E310 8E88 0004		00001088	863+	LG	R1,R1FUDGE	
00001496	B982 0022			864+	XGR	R2,R2	
0000149A	5820 5010		00000010	865+	L	R2,R2ADDR	load r2 source address
0000149E	E322 0000 0004		00000000	866+	LG	R2,0(R2)	load source
000014A4				869+	DS	0H	
000014A4	B9690012			870+	DC	XL4'B9690012'	CTZQ r1,r2
000014A8	E310 5028 0024		00001480	872+	STG	R1,R1010	save R1 output
000014AE	07FB			873+	BR	R11	return
000014B0				874+RE10	DC	0F	xl16 expected result
000014B0				875+	DROP	R5	
000014B0	00000000 00000006			876	DC	XL8'0000 0000 0000 0006'	result
000014B8	03555555 555555C0			877	DC	XL8'0355 5555 5555 55C0'	R2
				878			
				879	RRE	CTZG	
000014C0				880+	DS	0FD	
000014C0		000014C0		881+	USING	*,R5	base for test data and test routine
000014C0	000014F8			882+T11	DC	A(X11)	address of test routine
000014C4	000B			883+	DC	H'11'	test number

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000014C6	00			884+	DC	X'00'	
000014C7	C3E3E9C7 40404040			885+	DC	CL8'CTZG'	instruction name
000014D0	00001520			886+	DC	A(RE11+8)	address of R2 source
000014D4	00001528			887+	DC	A(RE11+16)	address of R3 source
000014D8	00000008			888+	DC	A(8)	result length
000014DC	00001518			889+REA11	DC	A(RE11)	result address
000014E0	00000000 00000000			890+	DS	FD	gap
000014E8	00000000 00000000			891+R1011	DS	D	R1 output
000014F0	00000000 00000000			892+	DS	FD	gap
				893+*			
000014F8				894+X11	DS	0F	
000014F8	E310 8E88 0004		00001088	895+	LG	R1,R1FUDGE	
000014FE	B982 0022			896+	XGR	R2,R2	
00001502	5820 5010		00000010	897+	L	R2,R2ADDR	load r2 source address
00001506	E322 0000 0004		00000000	898+	LG	R2,0(R2)	load source
0000150C				901+	DS	0H	
0000150C	B9690012			902+	DC	XL4'B9690012'	CTZQ r1,r2
00001510	E310 5028 0024		000014E8	904+	STG	R1,R1011	save R1 output
00001516	07FB			905+	BR	R11	return
00001518				906+RE11	DC	0F	xl16 expected result
00001518				907+	DROP	R5	
00001518	00000000 00000010			908	DC	XL8'0000 0000 0000 0010'	result
00001520	00005555 55550000			909	DC	XL8'0000 5555 5555 0000'	R2
				910			
				911	RRE	CTZG	
00001528				912+	DS	0FD	
00001528		00001528		913+	USING	*,R5	base for test data and test routine
00001528	00001560			914+T12	DC	A(X12)	address of test routine
0000152C	000C			915+	DC	H'12'	test number
0000152E	00			916+	DC	X'00'	
0000152F	C3E3E9C7 40404040			917+	DC	CL8'CTZG'	instruction name
00001538	00001588			918+	DC	A(RE12+8)	address of R2 source
0000153C	00001590			919+	DC	A(RE12+16)	address of R3 source
00001540	00000008			920+	DC	A(8)	result length
00001544	00001580			921+REA12	DC	A(RE12)	result address
00001548	00000000 00000000			922+	DS	FD	gap
00001550	00000000 00000000			923+R1012	DS	D	R1 output
00001558	00000000 00000000			924+	DS	FD	gap
				925+*			
00001560				926+X12	DS	0F	
00001560	E310 8E88 0004		00001088	927+	LG	R1,R1FUDGE	
00001566	B982 0022			928+	XGR	R2,R2	
0000156A	5820 5010		00000010	929+	L	R2,R2ADDR	load r2 source address
0000156E	E322 0000 0004		00000000	930+	LG	R2,0(R2)	load source
00001574				933+	DS	0H	
00001574	B9690012			934+	DC	XL4'B9690012'	CTZQ r1,r2
00001578	E310 5028 0024		00001550	936+	STG	R1,R1012	save R1 output
0000157E	07FB			937+	BR	R11	return
00001580				938+RE12	DC	0F	xl16 expected result
00001580				939+	DROP	R5	
00001580	00000000 00000018			940	DC	XL8'0000 0000 0000 0018'	result
00001588	000000AA 55000000			941	DC	XL8'0000 00AA 5500 0000'	R2

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				942			
				943	RRE	CTZG	
00001590				944+	DS	0FD	
00001590		00001590		945+	USING	*,R5	base for test data and test routine
00001590	000015C8			946+T13	DC	A(X13)	address of test routine
00001594	000D			947+	DC	H'13'	test number
00001596	00			948+	DC	X'00'	
00001597	C3E3E9C7 40404040			949+	DC	CL8'CTZG'	instruction name
000015A0	000015F0			950+	DC	A(RE13+8)	address of R2 source
000015A4	000015F8			951+	DC	A(RE13+16)	address of R3 source
000015A8	00000008			952+	DC	A(8)	result length
000015AC	000015E8			953+REA13	DC	A(RE13)	result address
000015B0	00000000 00000000			954+	DS	FD	gap
000015B8	00000000 00000000			955+R1013	DS	D	R1 output
000015C0	00000000 00000000			956+	DS	FD	gap
				957+*			
000015C8				958+X13	DS	0F	
000015C8	E310 8E88 0004		00001088	959+	LG	R1,R1FUDGE	
000015CE	B982 0022			960+	XGR	R2,R2	
000015D2	5820 5010		00000010	961+	L	R2,R2ADDR	load r2 source address
000015D6	E322 0000 0004		00000000	962+	LG	R2,0(R2)	load source
000015DC				965+	DS	0H	
000015DC	B9690012			966+	DC	XL4'B9690012'	CTZQ r1,r2
000015E0	E310 5028 0024		000015B8	968+	STG	R1,R1013	save R1 output
000015E6	07FB			969+	BR	R11	return
000015E8				970+RE13	DC	0F	xl16 expected result
000015E8				971+	DROP	R5	
000015E8	00000000 00000030			972	DC	XL8'0000 0000 0000 0030'	result
000015F0	05550000 00000000			973	DC	XL8'0555 0000 0000 0000'	R2
				974			
				975	RRE	CTZG	
000015F8				976+	DS	0FD	
000015F8		000015F8		977+	USING	*,R5	base for test data and test routine
000015F8	00001630			978+T14	DC	A(X14)	address of test routine
000015FC	000E			979+	DC	H'14'	test number
000015FE	00			980+	DC	X'00'	
000015FF	C3E3E9C7 40404040			981+	DC	CL8'CTZG'	instruction name
00001608	00001658			982+	DC	A(RE14+8)	address of R2 source
0000160C	00001660			983+	DC	A(RE14+16)	address of R3 source
00001610	00000008			984+	DC	A(8)	result length
00001614	00001650			985+REA14	DC	A(RE14)	result address
00001618	00000000 00000000			986+	DS	FD	gap
00001620	00000000 00000000			987+R1014	DS	D	R1 output
00001628	00000000 00000000			988+	DS	FD	gap
				989+*			
00001630				990+X14	DS	0F	
00001630	E310 8E88 0004		00001088	991+	LG	R1,R1FUDGE	
00001636	B982 0022			992+	XGR	R2,R2	
0000163A	5820 5010		00000010	993+	L	R2,R2ADDR	load r2 source address
0000163E	E322 0000 0004		00000000	994+	LG	R2,0(R2)	load source
00001644				997+	DS	0H	
00001644	B9690012			998+	DC	XL4'B9690012'	CTZQ r1,r2

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				1013 *
				1014 * table of pointers to individual load test
				1015 *
00001668				1016 M4TESTS DS 0F
				1017 PTTABLE
00001668				1018+TTABLE DS 0F
00001668	000010B0			1019+ DC A(T1)
0000166C	00001118			1020+ DC A(T2)
00001670	00001180			1021+ DC A(T3)
00001674	000011E8			1022+ DC A(T4)
00001678	00001250			1023+ DC A(T5)
0000167C	000012B8			1024+ DC A(T6)
00001680	00001320			1025+ DC A(T7)
00001684	00001388			1026+ DC A(T8)
00001688	000013F0			1027+ DC A(T9)
0000168C	00001458			1028+ DC A(T10)
00001690	000014C0			1029+ DC A(T11)
00001694	00001528			1030+ DC A(T12)
00001698	00001590			1031+ DC A(T13)
0000169C	000015F8			1032+ DC A(T14)
				1033+*
000016A0	00000000			1034+ DC A(0) END OF TABLE
000016A4	00000000			1035+ DC A(0)
				1036
000016A8	00000000			1037 DC F'0' END OF TABLE
000016AC	00000000			1038 DC F'0'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				1040	*****		
				1041	*Register equates		
				1042	*****		
		00000000	00000001	1044	R0	EQU	0
		00000001	00000001	1045	R1	EQU	1
		00000002	00000001	1046	R2	EQU	2
		00000003	00000001	1047	R3	EQU	3
		00000004	00000001	1048	R4	EQU	4
		00000005	00000001	1049	R5	EQU	5
		00000006	00000001	1050	R6	EQU	6
		00000007	00000001	1051	R7	EQU	7
		00000008	00000001	1052	R8	EQU	8
		00000009	00000001	1053	R9	EQU	9
		0000000A	00000001	1054	R10	EQU	10
		0000000B	00000001	1055	R11	EQU	11
		0000000C	00000001	1056	R12	EQU	12
		0000000D	00000001	1057	R13	EQU	13
		0000000E	00000001	1058	R14	EQU	14
		0000000F	00000001	1059	R15	EQU	15
				1061	*****		
				1062	*Register equates		
				1063	*****		
		00000000	00000001	1065	V0	EQU	0
		00000001	00000001	1066	V1	EQU	1
		00000002	00000001	1067	V2	EQU	2
		00000003	00000001	1068	V3	EQU	3
		00000004	00000001	1069	V4	EQU	4
		00000005	00000001	1070	V5	EQU	5
		00000006	00000001	1071	V6	EQU	6
		00000007	00000001	1072	V7	EQU	7
		00000008	00000001	1073	V8	EQU	8
		00000009	00000001	1074	V9	EQU	9
		0000000A	00000001	1075	V10	EQU	10
		0000000B	00000001	1076	V11	EQU	11
		0000000C	00000001	1077	V12	EQU	12
		0000000D	00000001	1078	V13	EQU	13
		0000000E	00000001	1079	V14	EQU	14
		0000000F	00000001	1080	V15	EQU	15
		00000010	00000001	1081	V16	EQU	16
		00000011	00000001	1082	V17	EQU	17
		00000012	00000001	1083	V18	EQU	18
		00000013	00000001	1084	V19	EQU	19
		00000014	00000001	1085	V20	EQU	20
		00000015	00000001	1086	V21	EQU	21

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
RELEN	A	00000018	4	453		
RPTDWSAV	D	00000390	8	326	313	317
RPTERROR	I	0000033C	4	299	274	
RPTSAVE	F	00000384	4	323	299	320
RPTSVR5	F	00000388	4	324	300	319
SKL0001	U	0000005E	1	219	235	
SKT0001	C	0000022A	20	216	219	236
SVOLDPSW	U	00000140	0	155		
T1	A	000010B0	4	558	1019	
T10	A	00001458	4	850	1028	
T11	A	000014C0	4	882	1029	
T12	A	00001528	4	914	1030	
T13	A	00001590	4	946	1031	
T14	A	000015F8	4	978	1032	
T2	A	00001118	4	590	1020	
T3	A	00001180	4	622	1021	
T4	A	000011E8	4	654	1022	
T5	A	00001250	4	686	1023	
T6	A	000012B8	4	718	1024	
T7	A	00001320	4	750	1025	
T8	A	00001388	4	786	1026	
T9	A	000013F0	4	818	1027	
TESTING	F	00001004	4	412	255	
TNUM	H	00000004	2	447	254	302
TSUB	A	00000000	4	446	258	
TTABLE	F	00001668	4	1018		
V0	U	00000000	1	1065		
V1	U	00000001	1	1066		
V10	U	0000000A	1	1075		
V11	U	0000000B	1	1076		
V12	U	0000000C	1	1077		
V13	U	0000000D	1	1078		
V14	U	0000000E	1	1079		
V15	U	0000000F	1	1080		
V16	U	00000010	1	1081		
V17	U	00000011	1	1082		
V18	U	00000012	1	1083		
V19	U	00000013	1	1084		
V2	U	00000002	1	1067		
V20	U	00000014	1	1085		
V21	U	00000015	1	1086		
V22	U	00000016	1	1087		
V23	U	00000017	1	1088		
V24	U	00000018	1	1089		
V25	U	00000019	1	1090		
V26	U	0000001A	1	1091		
V27	U	0000001B	1	1092		
V28	U	0000001C	1	1093		
V29	U	0000001D	1	1094		
V3	U	00000003	1	1068		
V30	U	0000001E	1	1095		
V31	U	0000001F	1	1096		
V4	U	00000004	1	1069		
V5	U	00000005	1	1070		
V6	U	00000006	1	1071		
V7	U	00000007	1	1072		

DESC	SYMBOL	SIZE	POS	ADDR
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Entry: 0

Image	IMAGE	5808	0000-16AF	0000-16AF
Region		5808	0000-16AF	0000-16AF
CSECT	MIE4TST	5808	0000-16AF	0000-16AF

STMT	FILE NAME
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```
1 /devstor/sharedvfp/tests/mie4-02-CLZG-CTZG.asm
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**** NO ERRORS FOUND ****